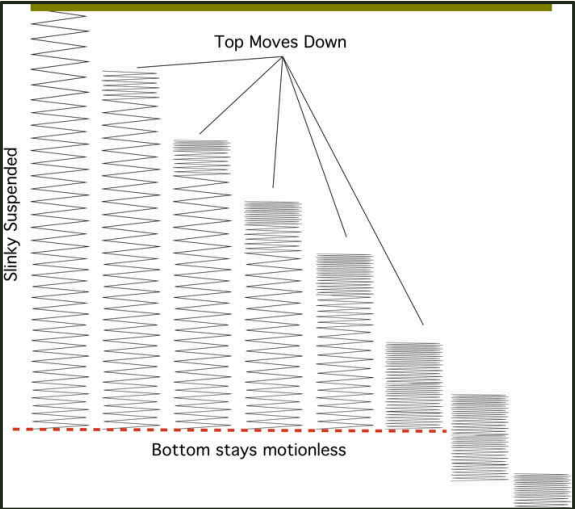
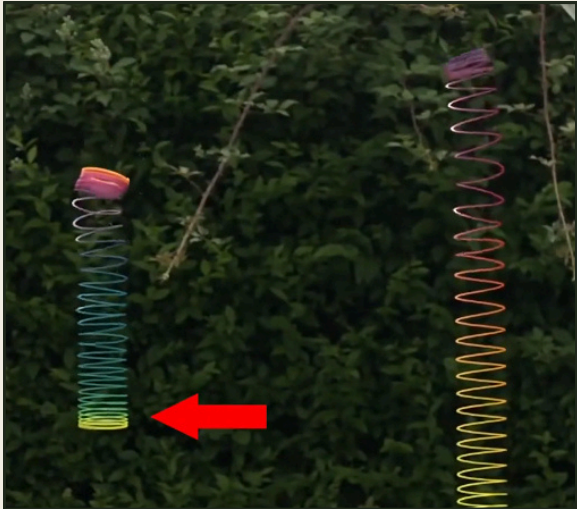


Slinky 101

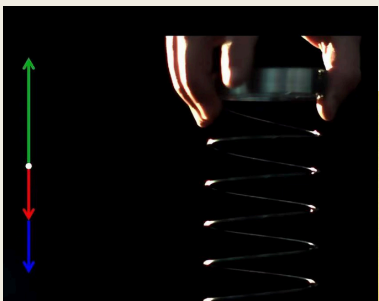
The Slinky Drop Experiment

Joel Ye, Tian Rongqi

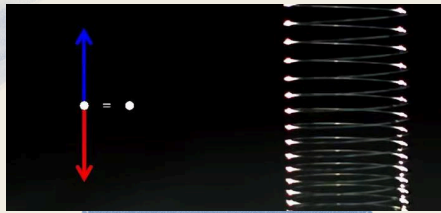


Explanation:

- The hand's upward force is transmitted down the Slinky as tension. It is pulled upward by the coil above it and downward by the coil below it
- Equilibrium exists because the upward pull from above on any coil = the weight of all the coils below it + the downward pull from the coil immediately below



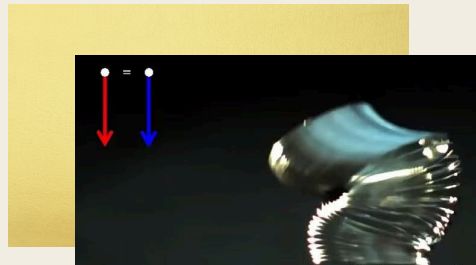
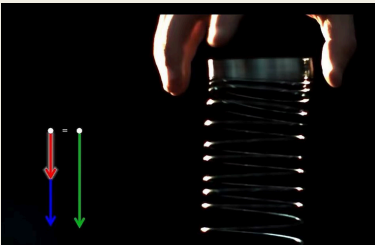
$$F_{hand} = mg + T$$



$$T = mg$$

- The tension in the Slinky decreases from top to bottom. The tension at any point = the weight of the segment hanging below it
- At the very top, tension = Slinky's total weight; At the very bottom, tension = ~0
- Weight on an individual coil is constant
- For the bottom-most coil, the upward tension from the coil above = its own downward weight, resulting in a net $F = 0$
- It is in a state of static equilibrium

- Releasing the top instantly removes the supporting tension for the top coil only
- The information "you are now unsupported" travels downward as a compression wave at a finite speed
- Coils ahead of this wavefront remain in their original static equilibrium. Therefore, the bottom coil does not move at all until the wave reaches it and destroys its balanced state



- Once the wave reaches the bottom coil, then the entire Slinky is in a state of free fall → it falls as a compact object
- The center of mass of the Slinky has been accelerating downward at g (9.8 m/s^2) from the very instant of release

M = mass
L = length
k = spring constant
 μ = linear density
v = wave speed

$$v = \sqrt{\frac{kL}{\mu}} = \sqrt{\frac{kL^2}{M}} \quad k = \frac{Mg}{L}$$

$$v = \sqrt{gL} \quad \mu = \frac{M}{L} \quad t_{collapse} = \frac{L}{v} = \sqrt{\frac{L}{g}}$$



It's all downhill form here



Kevin Tan, Jay Hari

Downward Motion:

- Governed by Newtonian Laws and wave motion
- Slinky moves in a transverse wave down the stairs
- Momentum carries the top down to the next step
- Continues through Potential Energy
- Time per step is dependent only on slinky length

$$t = \sqrt{2L/g}$$

- Highly complex equations of motion
 - Irregular center of mass movement

$$\ddot{y}_j + 4mR^2 \sum_{i=1}^N \left[\left(\sum_{k=1}^i \sin \theta_{k(-1)} \right)^2 \ddot{\theta}_k \right] \sum_{k=1}^i \sin \theta_{k(-1)} \ddot{\theta}_k + \left\{ \sum_{k=1}^i \cos \theta_{k(-1)} \ddot{\theta}_k \right\} \sum_{k=1}^i \cos \theta_{k(-1)} \ddot{\theta}_k - 4mR^2 \sum_{i=1}^N \left[\left(\sum_{k=1}^i \cos \theta_{k(-1)} \right)^2 \ddot{\theta}_k \right] \sum_{k=1}^i \cos \theta_{k(-1)} \ddot{\theta}_k + \left\{ \sum_{k=1}^i \sin \theta_{k(-1)} \ddot{\theta}_k \right\} \sum_{k=1}^i \sin \theta_{k(-1)} \ddot{\theta}_k \quad (13)$$
$$= 4mR^2 \sum_{i=1}^N \left\{ \sum_{k=1}^i \cos \theta_{k(-1)} \ddot{\theta}_k + \left(\cos \theta_{k(-1)} \right)^2 \ddot{\theta}_k \quad k \geq j \text{ otherwise} \right\}$$

- The world's longest slinky is 528m!

Well then can it go uphill??

- Not unless you push it
- Slinky is a torsional spring, twisting and untwisting about the position of equilibrium (consider these to be oscillations)
- Key idea is that the "arc" of the slinky gets smaller with each step climbed due to damping
- Light damping occurs due to:
 - Friction (with step)
 - Friction (internally)
 - Friction (air resistance)
- Intuitively: Loss in total KE + EPE → Lower total mechanical energy → GPE doesn't increase enough (Less energy than that required to lift centre of mass to next step)
- Oscillations become progressively smaller as amplitude decreases
- Just push it and it should go uphill just fine